Lesson for the course "Human-Computer Interaction"

Emotional Design

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Emotions and the user experience

- Emotions play a central role in the human ability to understand and learn about the world
  - Positive experiences stimulate our curiosity, and negative ones protect us from repeating mistakes

- HCI has traditionally been about designing efficient and effective systems

- Nowadays it is more about **how to design interactive systems that make people respond in certain ways**
  - e.g., to be happy, to be trusting, to learn, to be motivated…

Emotional interaction

- Is concerned with how we feel and react when interacting with technologies
- Then it uses this knowledge to inform the design of different aspects of the user experience
- It includes
  - **Emotional design**: how to elicit appropriate emotions, in order to create a positive experience for the user
  - **Affective computing**: development of methods to recognize and render emotions computationally
    - Area improving with better recognition software and machine learning algorithms

Emotional interaction

- Studies **what makes us** happy, sad, annoyed, anxious, frustrated, motivated and so on
  - And then translates this into different aspects of the user experience
  - E.g., why people become emotionally attached to certain products (e.g., virtual pets)
- This includes **how to change human behavior** through the use of emotional feedback
  - Interactive products can evoke certain kinds of emotional responses in people
  - Emotion drives behaviour…

Emotional design

- Emotional design strives to create products that elicit appropriate emotions, in order to create a positive experience for the user
- To do so, designers consider the connections that can form between users and the objects they use, and the emotions that can arise from them
- **The emotions a product elicits can <u>strongly</u> influence users' perceptions of it**

Emotional design

- To be an exceptional designer, it is not enough to just understand how your users are reacting – **you must understand why they react in that way**
  - The difficult task is having a deep understanding of your users' motivations and behaviour
  - Translating them into effective emotional design that is elegant, beautiful and truly unique will play a vital role in delivering an ideal user experience

Emotional System: 3-level model

- Don Norman proposes an emotional system consisting of 3 different, yet interconnected levels, each of which influences our experience of the world in a particular way
- **Visceral level** is responsible for the automatic and almost animalistic qualities of human emotion, which are almost entirely out of our control
- **Behavioral level** refers to the controlled aspects of human action, i.e., brain processes that control everyday behavior
  - E.g., well-learned routine operations such as talking, typing, and walking
- **Reflective level** is, as Don Norman states, *"…the home of reflection, of conscious thought, of learning of new concepts and generalisations about the world"*

Designing with the 3 levels in mind

- Humans form emotional connections with objects on the 3 levels, the visceral, behavioral, and reflective levels
- A designer should address the human cognitive ability at each level to elicit appropriate emotions so as to provide a positive experience
- We can **design products in terms of the 3 levels**
  - **Visceral design**
  - **Behavioral design**
  - **Reflective design**

Designing with the 3 levels in mind

- **Visceral design** appeals to our first reactions when we encounter a product
  - It mainly deals with aesthetics and the perceived quality from mere look and feel, and the engagement of the senses
  - Here we examine what "gut" reactions tell us about an item

- **Behavioral design** refers to the usability of the product, our assessment of how well it performs the desired functions, and how easily we can learn how to use it

Designing with the 3 levels in mind

- **Reflective design** is concerned with our ability to project the product's impact on our lives after we have used it
  - E.g., how it makes us feel when not holding it, or what values we find ourselves attaching to the product in retrospect
  - Here is where designers will want to maximize the users' desire to own that item

Designing with the 3 levels in mind



- Brilliant design: high on all 3 levels
  - Visceral (beauty)
  - Behavioral (usability)
  - Reflective (pride of ownership)



- Cultural images and graphical elements designed at reflective level
- Affordances of use at behavioral level
- Brilliant colors and wild design attract user's attention at visceral level

How to elicit emotions?

- There are a number of techniques
  - They depend on the specific context
  - You need to test them with users

- Examples include:
  - **Using color**: choosing the best combination of colors for an interactive design layout is crucial
  - **Making pages animated**
  - **Exploiting faults to trigger emotions** (such as the 404 error)

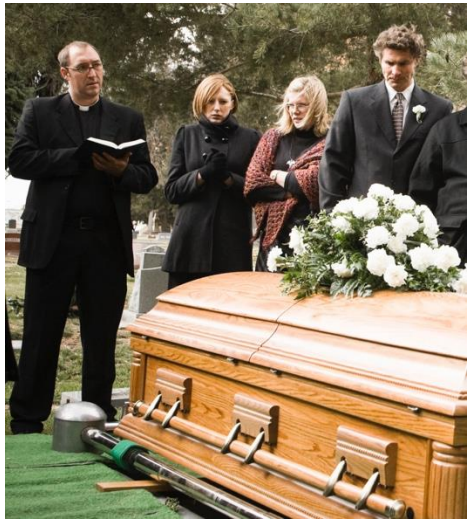
Using color to elicit emotions

- Colors can be used to **convey emotional content** as well as assist with the look and feel of your system/website/ /app/software/etc.
  - i.e., **to move people, evoke passions and feelings in our users**
- It's worth noting that people's **culture, gender, experiences**, etc. will also affect the way that colors resonate with them
- So, **user research is a better indicator** of emotional response to color than the following guidelines based on the "color wheel"

Using color to elicit emotions

- For instance, did you know that, in China, red is common because it represents happiness and prosperity, but white is considered funerary or representing misfortune?





- In Greece, yellow conveys notions of sadness, while red conveys such notions in South Africa
- Color is a big issue in how people from different parts of the world will interpret your design

Color temperature: evoking emotions



Color temperature

- **Warm colors**
  - These are colors located on the half of the color wheel that includes yellow, orange, and red
  - These colors are said to reflect feelings such as **passion, power, happiness, creativity, and energy**



Color temperature

- **Cool colors**
  - These are colors located on the other side of the color wheel, including green, blue, and purple
  - Cool colors are said to reflect **calmness, relaxation, meditation, freshness**



Color temperature

- **Neutral colors**
  - These colors include gray, brown, white, and black
  - These are **not said to reflect any particular emotions**



Color temperature: evoking emotions

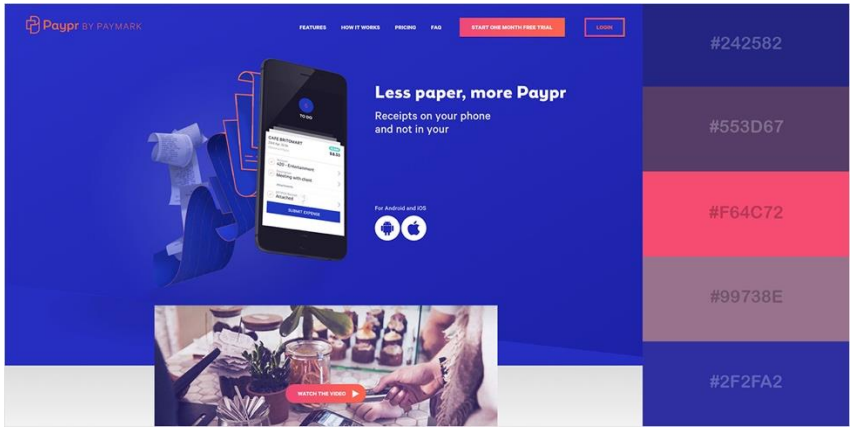
- While these groupings hold true **in a general sense**, emotional responses to colors may also be heavily affected by various factors (e.g., gender, experiences, cultural associations, other personal factors)
- You should **always test your color palettes** with your users to be sure that the choices you have made reflect their realities
  - It's almost always easier to set and test a color palette early in the development process than at the end

Suggested readings about meaning of colors

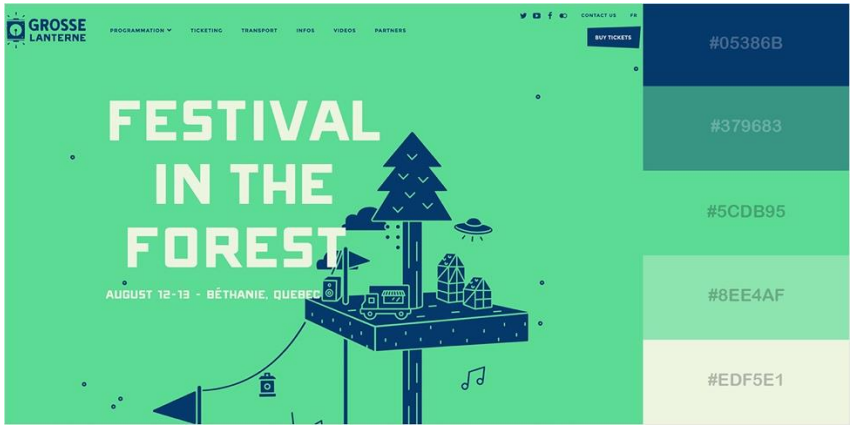
- How colors can influence people to make decisions and how the impact that colors can have on our brains
- https://uxdesign.cc/color-effect-b78fae8bb72f
- https://www.smashingmagazine.com/2010/01/color-theory-for-designers-part-1-the-meaning-of-color/

Find inspiration from good examples

- "50 Gorgeous Color Schemes From Award-Winning Websites"
  - https://visme.co/blog/website-color-schemes/



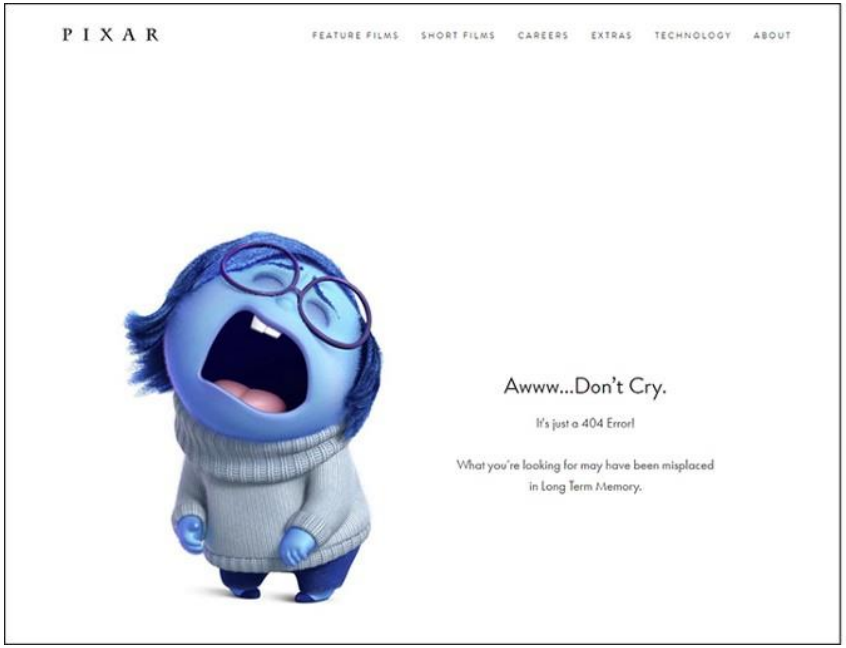


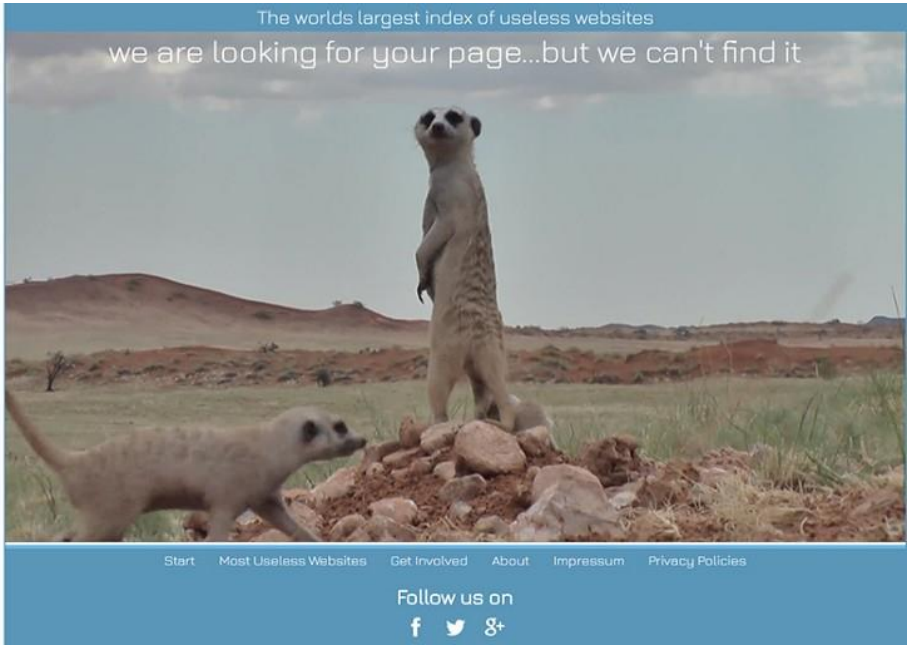




Example: the‘404 not found’ page

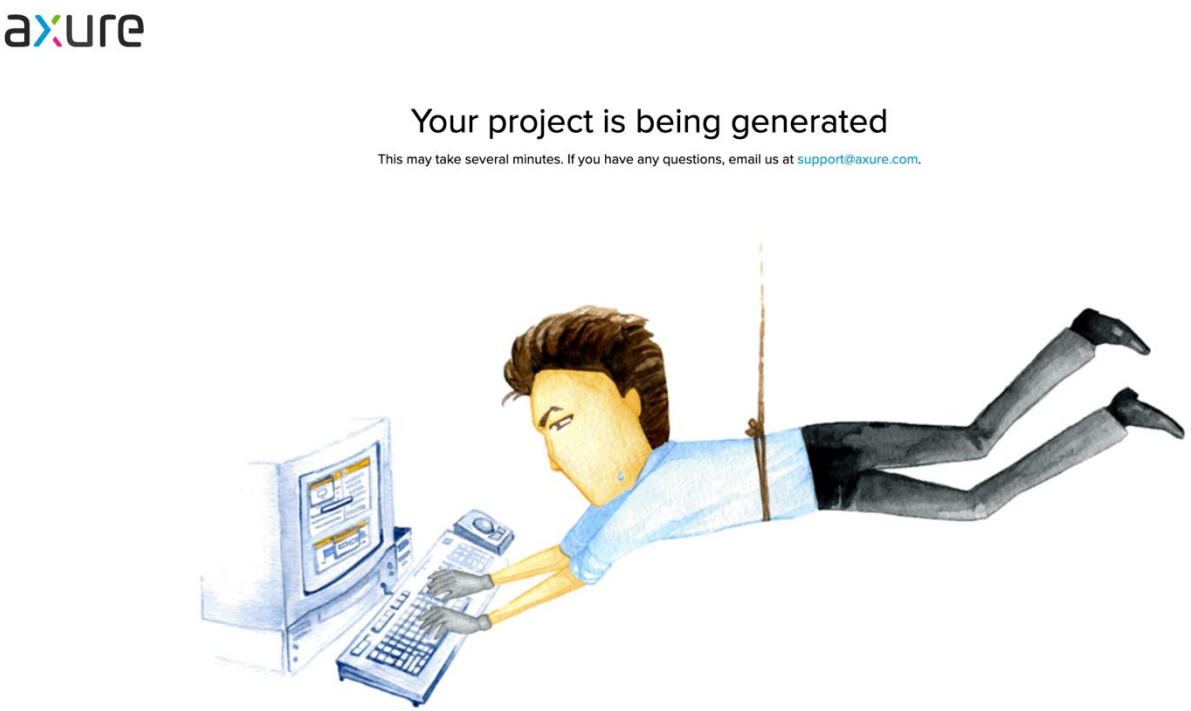
- **It is a good place to trigger emotions of your site visitors**
- Most users reaching the 404 error page will try to leave it as soon as possible
- Well-thought-out UX design of the 404 page will not only offer visitors links to relevant material, it will spark their interest and encourage to explore your offerings





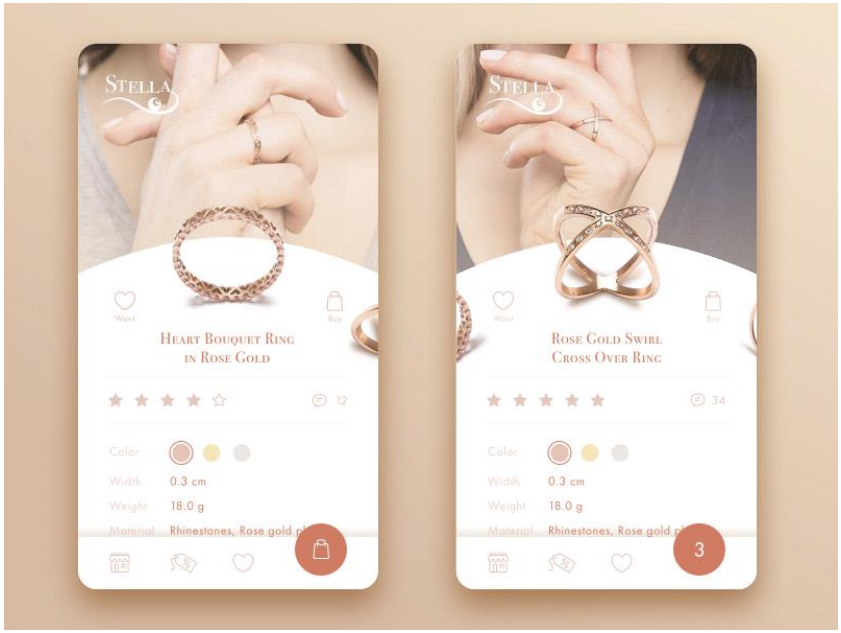
Example: waiting process to complete

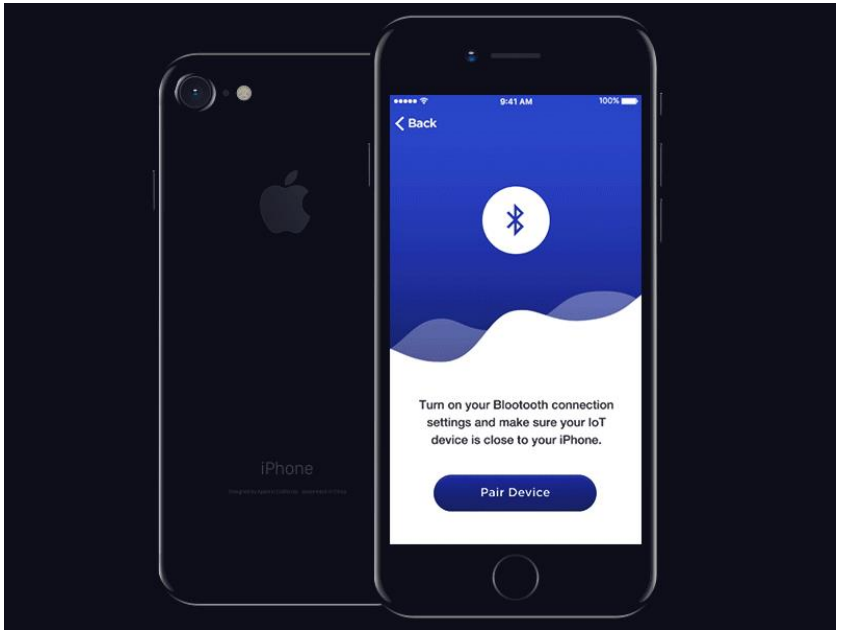
- It is a good place to trigger emotions of your site visitors **when they wait for a process to complete**



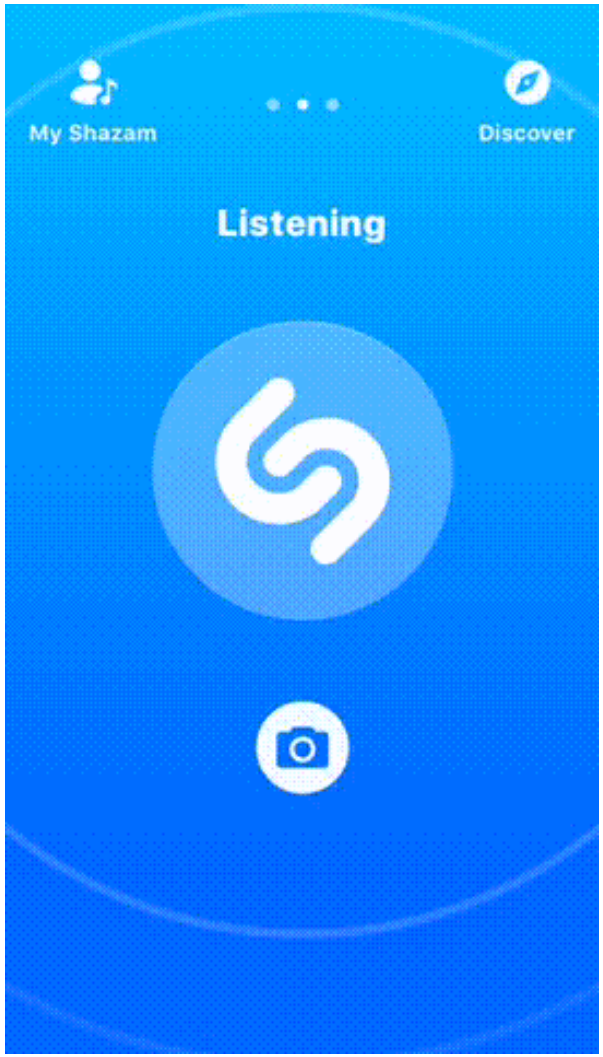
Animated user interfaces

- Increasing trend: animated micro-interactions and screen transformations to give the apps a "**personality**", to make them seem "**alive**"
  - The world around us doesn't operate with "hard-cuts." It's in motion, flowing and fluid; in constant transformation from one state to another
  - Animated user interfaces mimic the real-world and thus allow users to form a more human-like relationship with digital products





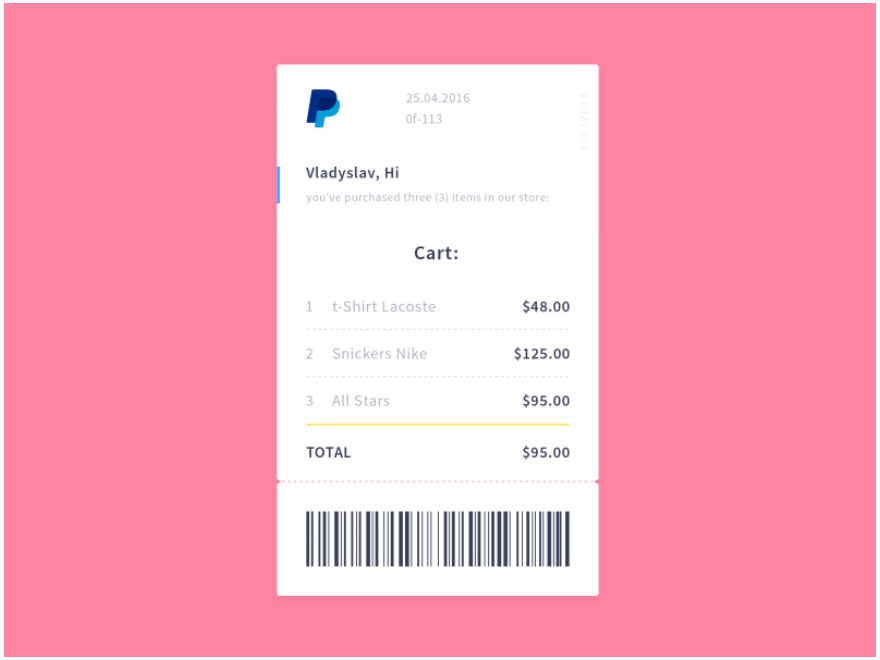
Example: Shazam



- When you use Shazam to find out what's playing, the button you tap starts sending concentric circles outward and inward
- This similarity to a throbbing audio speaker makes the interface almost tangible, physical — as if you're blasting your favorite album on a powerful sound system (metaphore)

Superfluous animations

- Animations are a crucial element of interaction design, **but they should always serve a purpose**
  - They should never be done for animation's sake
  - Unfortunately, designers tend to have a love affair with animations…



This animation is pretty, but superfluous. In total, it takes a whopping 3.5 seconds to see the transaction details. A simple fade-in of the receipt would be more elegant, and because it takes up less time, better for the user as well.

- **Always make your animation purposeful: too much can kill the UX of a product**

Negative emotions



- The experience of feeling emotions such as anger, frustration, guilt, nervousness and fear
- Inducing negative emotions in your users as they experience your user interface is one of main reasons why your product could fail in the market
- As a UX designer, you want users to enjoy using your product by eliminating negative experiences **and inducing positive emotion** in your users

Negative emotions

- Don Norman quote: *"Negative affect can make it harder to do even easy tasks: positive affect can make it easier to do difficult tasks"*
- How to prevent negative emotions in the user experience of your product?
- You can look to a variety of areas throughout the user experience **where you can induce positive emotion**
- The first step, however, is to learn to **identify parts of the user interface that could lead to negative emotion** in order to eliminate or fix those problem areas

Frustrating interfaces: causes

- When an application doesn't work properly or crashes
- When a system doesn't do what the user wants it to do
- When a user's expectations are not met
- When a system does not provide sufficient information to enable the user to know what to do
- When error messages pop up that are vague, obtuse or condemning
- When the appearance of an interface is garish, noisy or gimmicky
- When a system requires users to carry out too many steps to perform a task, only to discover a mistake was made earlier and they need to start all over again

Friction in UX hinders decision-making and leads to negative emotions

- The **abundance of choice** can be demotivating for people
  - When there is too much choice, people are less likely to decide on anything at all
  - And if they do decide on something, they end up feeling less satisfied with their selection

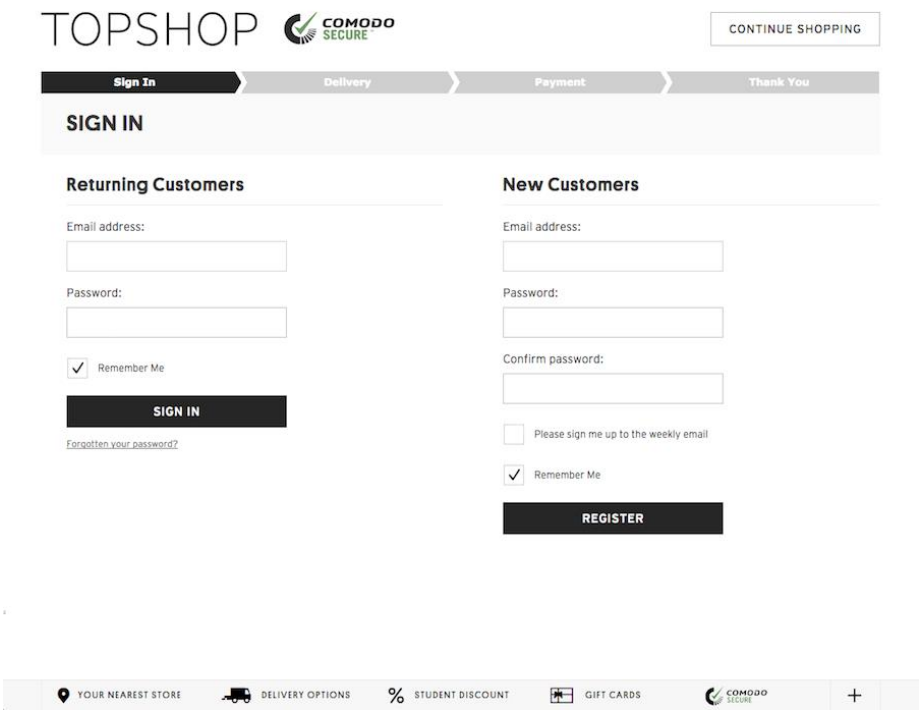


The overwhelming selection screen for the streaming site, Netflix, is an example of how an abundance of choice can cause users frustration and guilt due to indecision

Friction in UX hinders decision-making and leads to negative emotions

- Hindering your users from decision-making by offering too much choice causes negative emotions like frustration
- **Help users make decisions**
  - Make it easy for them to move through the user flow you've designed by aiding them to understand the concept or message you're trying to convey
  - …not only at every touch point, but also at any point in which the user lays eyes on the interface you've designed for them
  - Remember to provide the "information scent"

Avoid friction between users and the UI



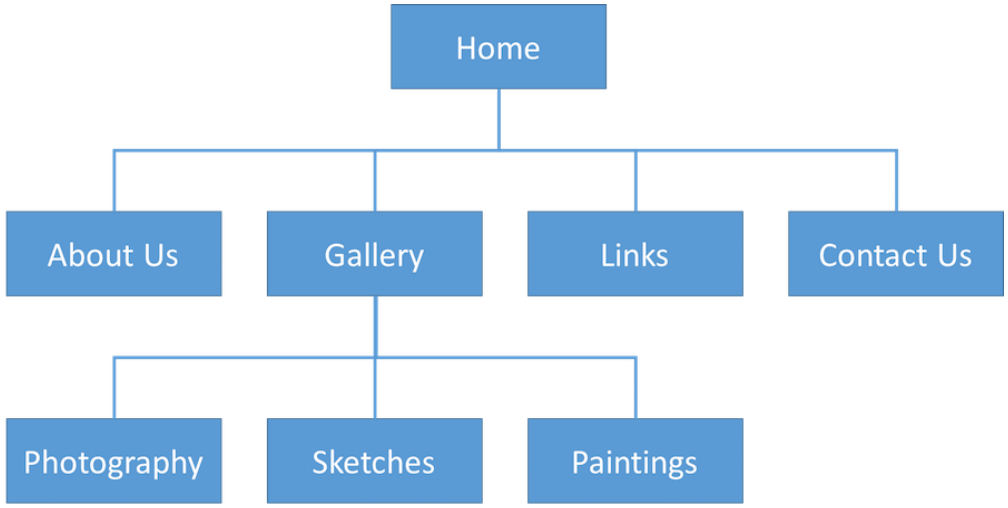
- When the users have decided to purchase and click checkout at this online shop they are led to an account registration or sign-in screen
- This extra step of "Create your account" or "Sign-in" before letting your users make a purchase can disrupt user flow causing them to feel frustration and annoyance and potentially even lose the sale
- Asking your users to create an account could be better included in a marketing campaign or something that shows it will benefit the user, not the other way around

Avoid friction between users and the UI

- Reduce friction between your users and the UI **by inducing "flow" state in your users**
  - Flow is the mental state a person goes through when they feel fully immersed when performing an activity
  - Immersion is often accompanied with a feeling of energized focus, full engagement, and enjoyment in the activity
- To achieve this, always seek to minimize disruption to your users by taking out any unpleasant experiences like pop up ads or forcing account logins
- This may be going against the demands of the marketing department, **but your job as a UX designer is to be an advocate for your users**
  - Therefore, fight for their benefit and explain to the team how each disruption you remove enhances the user experience

Avoid poor information architecture

- Information architecture is the structural foundation of any interface design
- Without it, your users will not be able to find what they're looking for, they will become frustrated, and you will lose their business
- Confusing navigation is often caused by poor placement of content in your UI as well as content hierarchy that is not very well thought out in the first place
- Creating a **proper site/app map** in the early stages can help you design a better navigation experience for your users in the end



Human decision making is rarely rational

- People are often irrational
  - We don't make decisions that are always in our best interests
  - Our unconscious governs a lot of our decisions
  - Harvard professor Gerald Zaltman says **95% of purchasing decisions are unconscious**
    - So….when marketing a product to a consumer, it is most effective to **target the unconscious mind**



We are biased by how choices are worded

- "The treatment has a 90% survival rate" vs "The treatment has a 10% mortality rate"
  - The two statements about the treatment's effectiveness are equivalent
  - A rational agent's decision would not be affected by how the doctor phrases it. But people's decisions are.

- **"Framing effect": how choices are framed affects people's decisions**

- Framing effects are one reason people's judgments and preferences are unstable over time
  - phrase a choice one way, people decide one way; phrase it differently, people decide differently

Exploiting strengths and weaknesses of human cognition

- We can design technology that:

1. **Supports rational decision making** → **Decision support systems**

2. **Leverages our unconscious** → **Persuasive systems**

Decision support systems

- Many software applications and commercial websites exist to help people make decisions
- Such websites **support rational choice** by displaying products side by side, allowing people to compare their price, features, and reliability, as well as customer satisfaction as indicated by product ratings and reviews
- E.g., Amazon (see next slide)

Decision support systems

Confronta con articoli simili

					
	Questo articolo TP-Link TL-WR802N Nano Router N300 Wi-Fi Portatile, 300 Mbps, 1 Porta LAN/WAN, 1 Porta Micro USB, Modalità Operative: Router, Repeater, Client, AP e WISP	GL.iNET GL-MT300N-V2 (Mango) Mini portabile router da viaggio VPN, Mobile Hotspot in tasca, ripetitore bridge, range extender, OpenVPN Client, 300Mbps Wireless ad alte prestazioni, 128MB RAM	Tenda 4G03 Router 4G LTE TE fino a 150 Mbps/Wireless fino a 300Mbps, Controllo Genitori, Monitoraggio Del Traffico Dati, Porta LAN / WAN, Senza configurazione, Accesso Wi-Fi Fino A 32 Dispositivi	STRONG Router 4G LTE WLAN 300M(LTE fino a 150 Mbit/S, 2.4 GHz WiFi @ 300 Mbit/S, 802.11b/g/N, porta LAN, adattatore SIM, bianco	TP-Link TL-MR6400 Router 4G LTE fino a 150 Mbps/Wireless N fino a 300Mbps, Porta LAN/WAN, Senza configurazione, Antenne Staccabili, Versione 5.0 il #1 più venduto
	Aggiungi al carrello	Aggiungi al carrello	Aggiungi al carrello	Aggiungi al carrello	Aggiungi al carrello
Valutazione cliente	★★★★½ (15595)	★★★★½ (6549)	★★★★½ (1922)	★★★★☆ (34)	★★★★½ (11274)
Prezzo	22,99 €	28,90 €	39,99 €	45,99 €	57,90 €
Spedizione	✓prime	✓prime	✓prime	✓prime	✓prime
Venduto da	Amazon.it	GL.iNet Technologie	Amazon.it	Amazon.it	Amazon.it
Tecnologia di connettività	Ethernet, Wi-Fi	Wi-Fi, USB, Ethernet	Wi-Fi, Ethernet	Ethernet, WLAN	Wi-Fi, Ethernet
Velocità di trasferimento dati	300 Mb al secondo	300 MB al secondo	150 Mb al secondo	300 Mb al secondo	150 Mb al secondo
Descrizione dell'interfaccia di rete	Ethernet	—	—	—	—
Standard per comunicazioni wireless	802.11a/b/g/n	802.11bgn	802.11b, 802.11g, 802.11n	802.11g, 802.11n, 802.11b	802.11b, 802.11g, 802.11n

Screenshot

Decision support systems: guidelines

- **Provide all options**
  - If there are too many to list in a simple way, then organize or abstract them into categories and subcategories, and provide summary information for those
  - …so people can evaluate and compare entire categories of options at once
- **Help people find alternatives**
  - Some solutions may be so counterintuitive that people don't consider them
  - Decision support systems can expose options users might miss and can generate variants on user solutions that are minor or even major improvements

Decision support systems: guidelines

- **Provide unbiased data**
  - i.e., data created or collected in an objective, reproducible manner
- **Don't make people calculate**
  - Perform calculations, inferences, and deductions for users where possible
  - Computers are good at that; people aren't

Persuasive systems

- **How an interactive system presents information influences the decisions and behavior** of users at least as strongly as what information it presents

- If designers want to influence or persuade people to respond in a specific way – for example, buy a product, join an organization, subscribe to a service, donate to a charity, form a certain political opinion, vote a certain way – they can "exploit" the characteristics of our unconscious to achieve that



Persuasive systems

- Advertising agencies have developed a number of techniques to influence people's emotions
- E.g., showing a picture of a cute animal or a child with hungry eyes on a website that "pulls at the heartstrings"
  - The goal is **to make people feel sad or upset** at what they observe and make them want to do something to help, such as by making a donation

Persuasive systems



Example of a web page designed to trigger a strong emotional response in the viewer

Persuasive systems

- Nowadays, there are many kinds of mobile **apps** and personal tracking **devices** available that are intended to help people monitor various behaviors and change them based on the data collected and displayed back to them
  - These devices include fitness trackers (e.g., Fitbit) and weight trackers (such as smart scales)
- The darker side: deceptive technologies
  - Phishing scams: there are still many vulnerable individuals



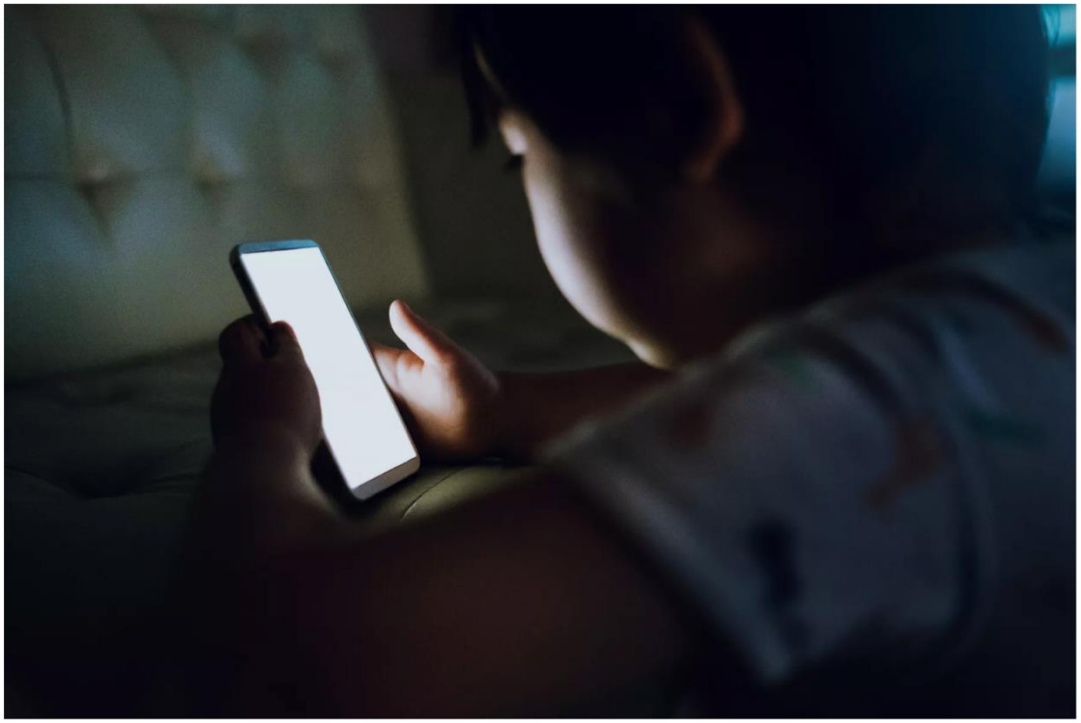
Persuasive systems

Tech companies use “persuasive design” to get us hooked. Psychologists say it’s unethical.

A group of psychologists say kids are suffering from “hidden manipulation techniques” that companies like Facebook and Twitter use.

By Chavie Lieber | @ChavieLieber | Chavie.Lieber@Vox.com | Aug 8, 2018, 2:30pm EDT

SHARE



Kids now have 10 times the amount of screen time they did in 2011, according to one study. | Getty Images

AD

Ad closed by Googl

Affective computing

- **Affective computing** is concerned with how to use computers to **recognize** and **express** emotions as humans do
  - Involves designing ways for people to communicate their emotional state
  - Uses sensing technologies to measure galvanic skin response, facial expressions, gestures, body movement



Emotional AI

- **Emotional AI** aims to automate the measurement of feelings and behavior using AI to infer them from facial expressions, voice, etc.
  - Aims to **predict** user's emotions and aspects of their behavior
  - E.g., *what is someone most likely to buy online when feeling sad, bored or happy*
- Technique used include
  - Cameras for measuring facial expressions
  - Biosensors placed on fingers or palms to measure GSR
  - Affective expression in speech (e.g. intonation, pitch, loudness)
  - Body movement and gestures using accelerometers and motion capture systems

Facial coding using Affdex software

- Six core expressions typically measured:
  - Sadness
  - Disgust
  - Fear
  - Anger
  - Contempt
  - Joy



- Type of facial expression chosen by AI through detecting presence/absence of:
  - Smiling, eye widening, brow raising, brow furrowing, raising a cheek, mouth opening, upper-lip raising, wrinkling of the nose

- Check the website https://www.affectiva.com

How can this emotional data be used?

- Ads in websites
  - If user screws up their face when an ad pops up -> feel disgust
  - If start smiling -> they are feeling happy
  - A website/app can adapt its ad, movie storyline or content to match user's emotional state
- If the system is used in a car it might detect an angry driver and suggests that the driver takes a deep breath
- Eye-tracking, finger pulse, speech and words/phrases can be analyzed when tweeting or posting to Facebook
- *Do you think it is **ethical** that technology can read your emotions from your facial expressions or from your tweets?*

Tips for the exam project

- You can add emotional components in your medium fidelity prototype (colors, images, system to recognize emotions, etc.): say it in the report!
- Ask yourself what is the goal of your design in terms of recognizing/eliciting certain emotions during the various steps of the user interaction with the system
- Remember to test with users the emotional impact of your design
  - Observe
  - Ask questions (small interview about the experience after the testing session of the prototype)
  - Use the Self-Assessment Manikin technique in questionnaires

Lesson for the course "Human-Computer Interaction"

Google Material Design (Part 4)

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Launch screen and empty states

- They are a good place to trigger emotions of your site visitors
  - https://material.io/design/communication/launch-screen.html
  - https://material.io/design/communication/empty-states.html

Homework

- Choose a website and an app and for each page assess what emotions it communicates to you:
  - Positive or negative?
  - Which emotion exactly?
- Analyze the colors of a website and an app: what emotions do they communicate to you?
- Select a website and check if there is an abundance of choice, the presence of tech-speak, or a poor information architecture

Homework

- Read the following Google's Material Design Guidelines:
  - https://material.io/design/communication/launch-screen.html
  - https://material.io/design/communication/empty-states.html

- Choose one of the Material Design case studies and have a look at it:
  - https://material.io/design/material-studies/about-our-material-studies.html

Bibliography

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